

FINAL REPORT
General Motors Project
Group 8



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Executive Overview

General Motors (GM) is one of the world's largest automotive manufacturers, with a strong presence in electric vehicle (EV) innovation and sustainability leadership. GM's vision is rooted in achieving a future of zero crashes, zero emissions, and zero congestion. With brands like Chevrolet, Cadillac, GMC, and Buick under its umbrella, GM has been aggressively expanding its electric vehicle lineup and associated infrastructure. This strategic vision is not only about manufacturing EVs but also about enabling the ecosystem that supports long-term adoption and accessibility across diverse regions of the United States.

In pursuit of this mission, GM partnered with our Capstone team to address a key infrastructure gap: the strategic placement of 100 new DC fast chargers in Tennessee. Tennessee presents a unique opportunity due to its combination of strong EV growth potential, demographic diversity, and the presence of GM's manufacturing plant in Spring Hill. However, many regions within the state either lack charging infrastructure altogether or have inefficiently distributed chargers that don't align with actual or projected demand. The challenge was to identify ZIP codes where public charger deployment would yield the greatest return in terms of utilization, accessibility, and EV adoption.

To solve this, our team conducted extensive data analysis using a combination of exploratory data analysis (EDA), regression modeling, and greedy optimization algorithms. We integrated multi-source data—including EV registrations, demographic indicators, traffic patterns, business density, and existing charger locations—to build a decision-support framework. The outcome is a prioritized deployment strategy, supported by an interactive Tableau dashboard, that enables GM to efficiently roll out chargers in underserved yet high-potential ZIP codes, ultimately driving EV sales, customer satisfaction, and sustainability impact in the region.

Articulation of Business Outcomes

Business Context

Tennessee presents a compelling business case for electric vehicle (EV) infrastructure investment, particularly for General Motors (GM), which maintains a strong presence in

the state through its Spring Hill manufacturing facility. Despite increasing EV registrations and consumer interest, the distribution of public EV chargers across the state is uneven and misaligned with demand. Urban areas and high-income ZIP codes are often well-equipped, while suburban and rural corridors—many with growing populations, high commute times, and positive demographic signals—remain underserved. This infrastructure gap limits the effectiveness of GM's EV rollout strategy, creates friction in customer experience, and under-utilizes the full potential of high-growth regions. Given GM's strategic vision of zero crashes, zero emissions, and zero congestion, the need to address these imbalances is both urgent and aligned with corporate priorities. Our project aims to equip GM with a robust, data-driven framework to identify and prioritize ZIP codes that maximize EV adoption and charger utilization. By doing so, we help GM optimize its infrastructure investments while reinforcing its position as a leader in sustainable mobility.

Business Questions

In addressing the deployment of 100 new DC fast chargers across Tennessee, several business-critical questions emerged that guided our analytical approach. First and foremost, GM sought to understand where these chargers should be placed to drive the highest return on infrastructure investment. This meant identifying not just high-population areas but ZIP codes with the right combination of current and forecasted EV registrations, traffic volumes, demographic readiness, and absence of sufficient public charging infrastructure. Additionally, GM needed to know what factors influence EV adoption the most—whether charger availability alone drives behavior, or whether socioeconomic variables such as income levels, education attainment, and commute patterns play a larger role in EV market penetration. Beyond these predictive questions, the project also explored how to balance equity and efficiency, ensuring that underserved communities are not left behind while still maximizing charger utilization. Finally, GM was interested in understanding the scalability of the model—could this framework work not just in Tennessee, but across other states with different urban-rural dynamics and EV readiness levels? These questions framed every decision, from data selection to model evaluation, and helped ensure that our solution was both strategically sound and operationally viable.

Desired Outcomes

The primary outcome of this project is to deliver a strategic, data-backed deployment plan for 100 new DC fast chargers in Tennessee that aligns with GM's broader EV infrastructure goals. This involves more than just placing chargers where there is high population—it means ensuring that charger placement directly supports increased EV sales, improves customer convenience, and reinforces GM's role as a leader in

sustainable transportation. By combining demographic analysis, EV registration trends, business and traffic density, and infrastructure availability, the model identifies 74 ZIP codes where investment is most likely to yield strong utilization and support long-term growth. Collectively, these locations are projected to fulfill 52.6% of Tennessee's 2028 EV charging demand, making the plan not only impactful but also future-ready. These outcomes are crucial for improving the ROI of infrastructure investments and maximizing the positive externalities of EV adoption, such as reduced emissions and congestion. Additionally, the deployment plan supports GM's ambition to lead the national EV movement by showing a replicable model that can be extended beyond Tennessee. Ultimately, our work delivers a tangible strategy that bridges the gap between market demand, policy incentives, and infrastructure readiness, ensuring GM remains at the forefront of the EV transition.

Root Cause Analysis

To understand the existing gaps in EV charging infrastructure across Tennessee, our team began with a comprehensive root cause analysis informed by exploratory data analysis (EDA) and stakeholder discussions. We observed that although EV registrations were growing rapidly in several ZIP codes, the presence of public chargers did not reflect this demand. Notably, over 350 ZIP codes had little to no public charging infrastructure despite demonstrating high potential based on demographic and behavioral indicators. In contrast, some areas with relatively low EV growth were oversupplied with chargers. This misalignment suggested a lack of standardized, data-driven methods guiding charger placement historically.

Through geospatial analysis and demand-capacity mapping, we identified several contributing factors. First, infrastructure expansion had been reactive rather than strategic—often focused on urban centers or highways without considering future growth areas such as emerging suburban corridors. Second, public policy and investment incentives tended to prioritize visibility over actual usage potential, leading to underutilized chargers in well-connected zones and unmet demand in secondary markets. Third, the absence of a unified dataset combining charger locations, EV registrations, and local socio-economic data limited GM and other stakeholders from optimizing placement for maximum ROI. Finally, we found inconsistencies in how charger needs were evaluated, with no formal weighting of key variables such as population density, household income, education level, commute time, or traffic flow.

These findings made it clear that an updated, analytically rigorous approach was essential. To support this, we curated and merged a wide range of datasets at the ZIP-code level to form a unified analytical base. These included EV registration data from 2020 to 2024 to capture growth trends, charger location data for both DC fast and Level 2 chargers, and demographic profiles such as income levels, education attainment, household composition, and commute times. In addition, we leveraged business density data using Google Places API, traffic data from state-level transportation datasets, and rental estimates to project cost implications by location type. By standardizing, cleaning, and enriching these datasets, we created a robust foundation that informed our EDA, regression analysis, and final optimization logic—ensuring each ZIP code was evaluated not only on need but also on its strategic fit for deployment.

The lack of a unified prioritization framework resulted in inefficiencies both in capital deployment and customer experience. Our model was designed to address these root causes by unifying multi-source data, applying statistical rigor, and introducing a scalable decision-making tool that GM can use for this and future infrastructure rollouts.

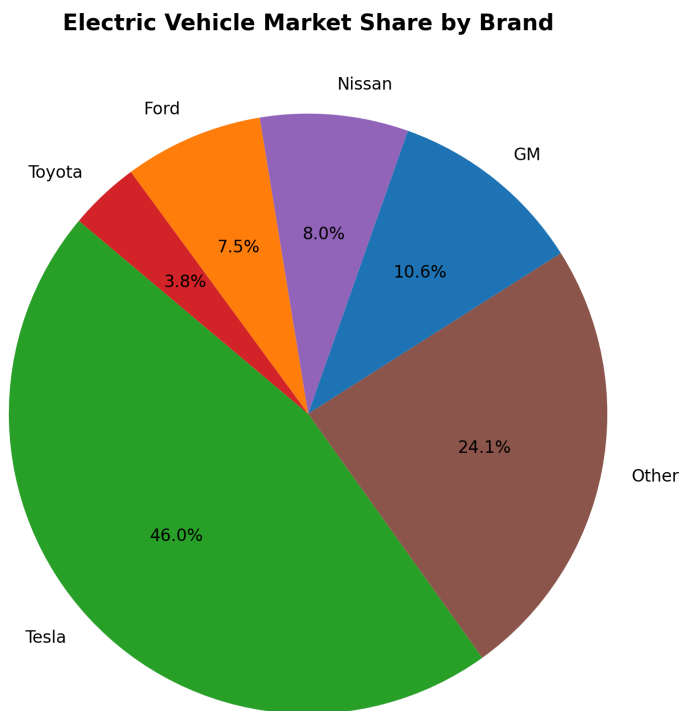
Data Exploration

We merged over five datasets at the ZIP-code level to inform our analysis, including:

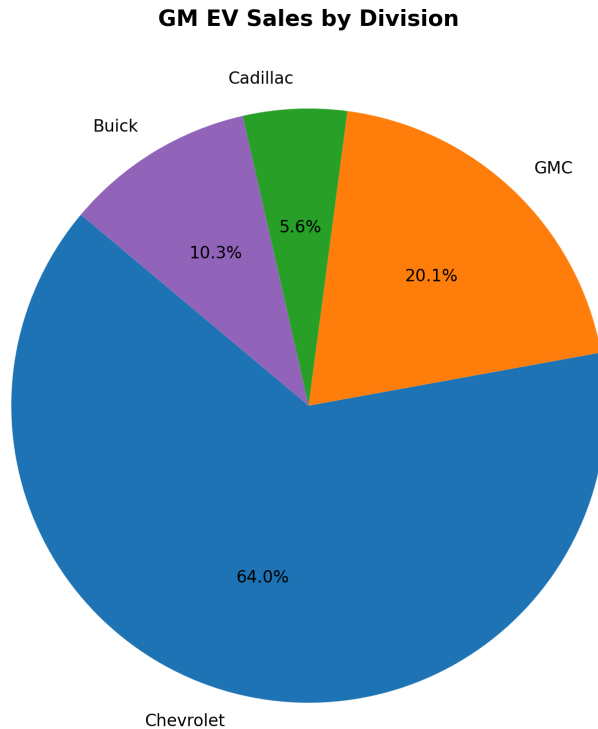
- EV registrations from 2020–2024
- Public charger availability, broken down by DC fast and Level 2 stations
- U.S. Census and ACS demographic indicators such as household income, education levels, population under 10 years old, and average commute times
- Google Places API-based business density for measuring commercial activity
- State-level traffic counts and inferred vehicle usage intensity
- Rent estimates for cost modeling in infrastructure ROI calculations

Exploratory Data Analysis (EDA) revealed significant disparities in how chargers were distributed across Tennessee. High-income ZIP codes often had greater charger availability, while emerging suburban areas with high EV demand indicators remained underserved. We discovered that education level, family composition, and commute times correlated strongly with higher EV adoption. A major insight from the data was that over 350 ZIP codes had no chargers at all, yet many of these ZIPs showed clear signs of demand potential.

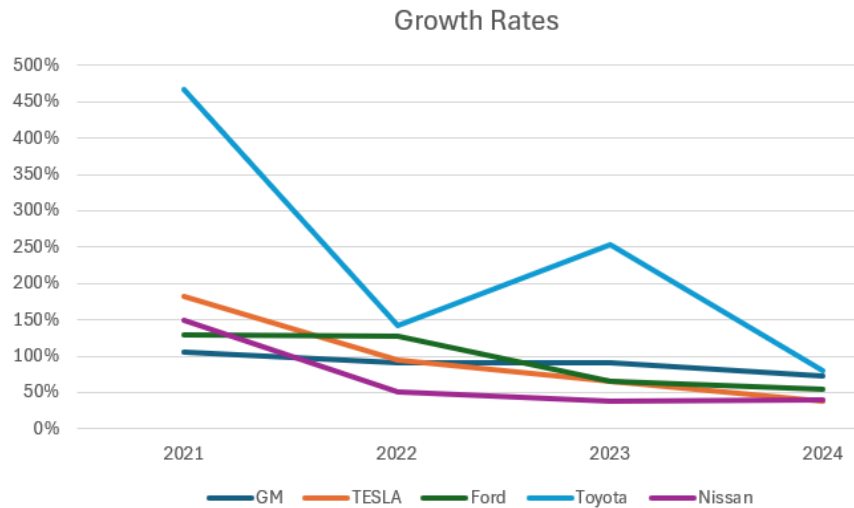
To contextualize GM's position in the broader EV landscape, we analyzed market share and sales trends.



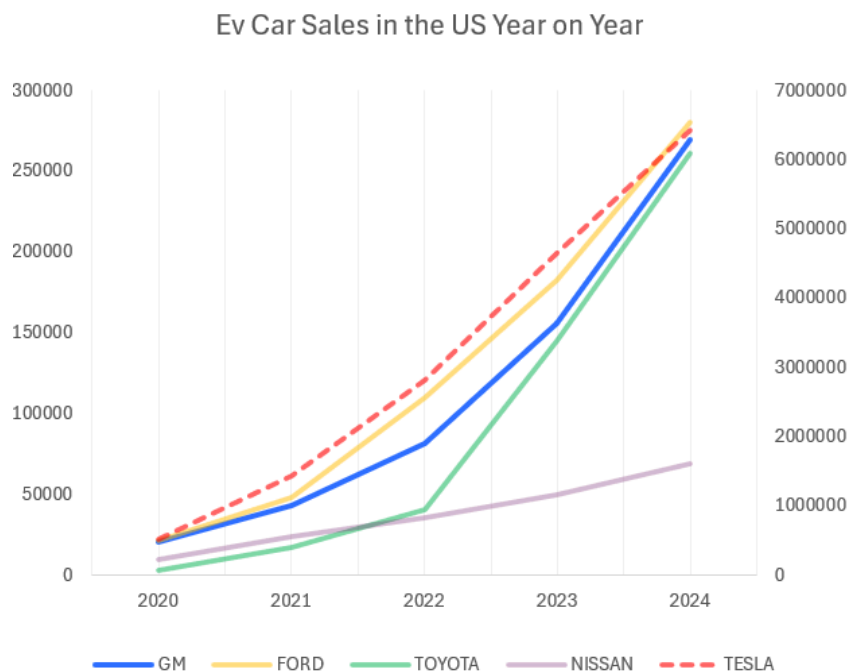
This pie chart illustrates the competitive landscape of electric vehicle (EV) manufacturers in the U.S. market. Tesla dominates the EV space with a commanding 46.0% market share, reinforcing its position as the industry leader. The next largest segment, labeled "Other," comprises 24.1%, indicating a fragmented field of smaller brands. GM holds a 10.6% share, placing it as a significant but still trailing player in the EV segment compared to Tesla. Other major automakers include Nissan (8.0%), Ford (7.5%), and Toyota (3.8%). The chart highlights the pressing need for GM to accelerate its infrastructure and product strategies to gain greater market penetration and compete with Tesla's first-mover advantage.



This chart breaks down GM's internal EV market share by division, shedding light on which brands contribute most to its EV sales. Chevrolet leads overwhelmingly with 64.0% of GM's EV sales, reflecting the brand's broad market appeal and affordability compared to premium GM lines. GMC follows with 20.1%, while Buick and Cadillac represent 10.3% and 5.6%, respectively. This distribution implies that any infrastructure planning—like charger placement—should consider Chevrolet's dominant customer base, as they are the most likely users of public charging stations. It also suggests potential for growth in GM's premium divisions as infrastructure expands and consumer segments diversify.



This line graph displays year-over-year growth rates in EV sales from 2021 to 2024 for major manufacturers. Tesla shows the most dramatic fluctuations, peaking with a 450% growth rate in 2021 and then normalizing in subsequent years. GM exhibits steady, though less volatile, growth, suggesting a more gradual market ramp-up. Other brands like Ford, Toyota, and Nissan show modest and more stable growth patterns. The overall trend reflects an early spike in EV interest followed by sustained but more tempered expansion, underscoring the importance of infrastructure readiness to support a growing but maturing market.

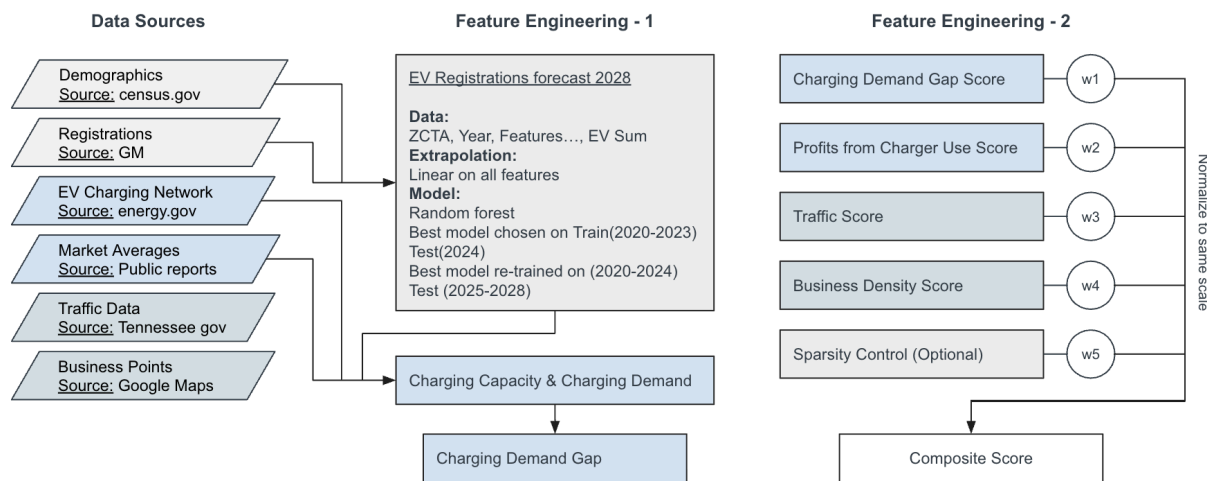


This graph charts the actual number of EVs sold by major automakers from 2020 through 2024. Tesla leads consistently in total unit sales, with a steep growth trajectory that outpaces all competitors. GM shows a strong upward trend, gradually closing the gap and indicating successful market traction. Ford and Toyota exhibit moderate but increasing sales volumes, while Nissan lags with slower growth. The graph validates the surge in consumer adoption of EVs and reinforces the urgency for parallel growth in public charging infrastructure to sustain and amplify these adoption trends. For GM, the data justifies investment in charging stations to maintain its positive momentum.

Model-Based Analysis

Our model-based approach was designed to bridge raw data insights with practical deployment strategies. It involved a multi-stage pipeline that systematically translated real-world variables into optimized charger placements. This process was structured around three key elements presented in the following slides.

Project Workflow – Data



This outlines our feature engineering process. We consolidated data from six primary sources—demographics, EV registrations, existing charging infrastructure, business density, traffic volumes, and market averages—to derive base features. A Random Forest model was trained on historical EV data to forecast 2028 registrations. This

output, paired with assumptions on usage patterns, was used to calculate both charging demand and capacity, which laid the foundation for prioritization scores.

Metric Calculations

Building on the forecast and capacity metrics, we developed five standardized indicators per ZIP code: Charging Demand Gap, Profits from Charger Use, Traffic Score, Business Density Score, and Sparsity Control. These scores incorporated assumptions around EV driving behavior, revenue and cost estimates, and exposure metrics. They were then normalized and weighted to produce a single composite score for each ZIP code. Below are the detailed calculations for each of our indicators:

Charging Demand Gap: Charging Demand - Charging Capacity

$$\text{Charging Demand} = \text{EV_Cars_2028} * \text{Avg_miles_driven_per_car} * \text{perc_charged_in_public} \\ \text{Count_per_zipcode} * 42.4 * 36.8\%$$

vehicle	miles	share	public chg use	equiv. miles
all	42.4	14%	100%	5.936
some	42.4	45.6%	50%	9.6672
none	42.4	40.4%	0%	0
				15.6032
home charging available		86%		
use public chargers		59.6%		
				36.8%

$$\text{Charging Capacity} = ((\text{DC_chrgr_cnt} * \text{Avg_range_per_hour}) + (\text{L2_chrgr_cnt} * \\ \text{Avg_range_per_hour})) * \text{hours_in_operation} \\ (\text{DC_count_per_zipcode} * 180) + (\text{L2_count_per_zipcode} * 25) * 14$$

Profits from charger use:

$$\text{DC_capacity_per_charger} * \text{sell_price_per_mile} - (\text{DC_capacity_per_charger} * \\ \text{cost_price_per_mile} + \text{rent_per_charger/day}) \\ \text{DC_count_per_zipcode} * 180 * \$0.14 - (\text{L2_count_per_zipcode} * 25 * \$0.04 + \\ \text{rent_per_zipcode_classification})$$

Sparsity control (optional, kept at 0 weight for our solution):

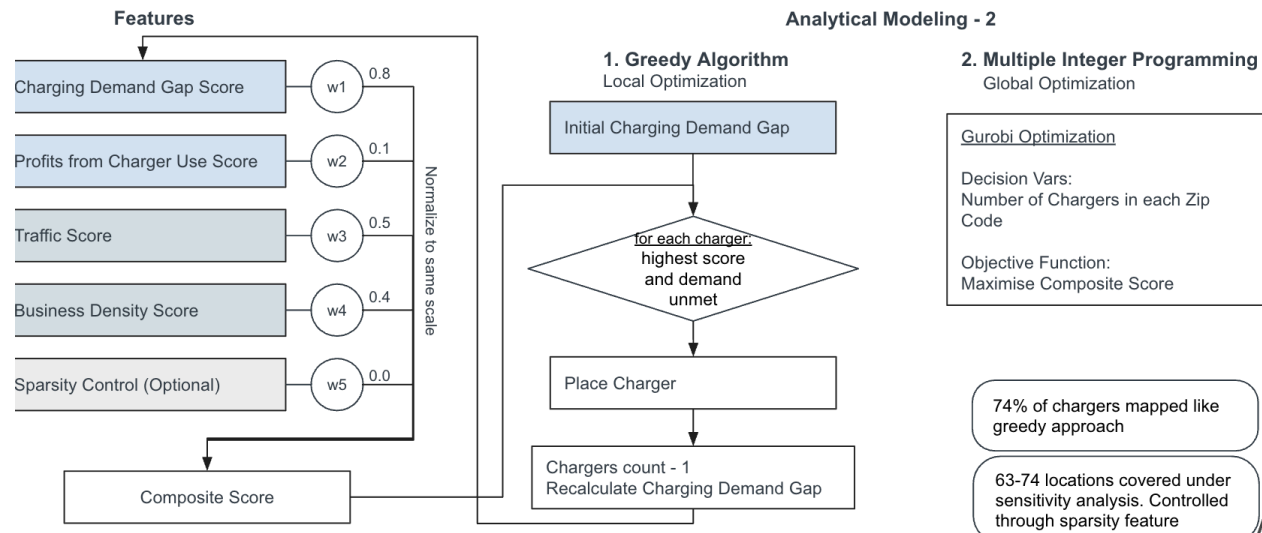
$$(\text{Feature1_std} * \text{Weight of Coeff 1}) + (\text{Feature2_std} * \text{Weight of Coeff 2}) + \dots + (\text{FeatureN_std} \\ * \text{Weight of Coeff N})$$

//Features selected by running regression analysis on ev car sales with demographic factors, keeping only statistically significant ones

Traffic Data (VMT; Vehicle Miles Travelled):

Annual sum of miles traveled by all vehicles within a zip code

Project Workflow – Charger Placement



Analytical Methods Tried

To optimize charger placement, we developed a two-phase modeling framework combining local and global optimization strategies. Central to both approaches was a composite score, calculated using five normalized features: charging demand gap, projected profits from charger use, traffic flow, business density, and optional sparsity control. These features were weighted according to stakeholder priorities, with demand gap carrying the highest influence (weight: 0.8). The other scores—traffic (0.5), business density (0.4), and charger-use profit (0.1)—provided additional context on usage likelihood and economic return.

We first implemented a Greedy Optimization Algorithm, which placed one charger at a time in the ZIP code with the highest unmet composite score. The demand gap was then updated after each placement, recalculating scores iteratively to reflect dynamic changes. This method allowed rapid scenario testing and transparent, interpretable results.

To validate these results, we implemented a Mixed Integer Programming (MIP) model using Gurobi. The global optimization model used integer decision variables to assign chargers across ZIP codes, with the objective of maximizing the total composite score under constraint. By comparing both approaches, we found that 74% of the placements overlapped, confirming that the greedy model achieved near-optimal outcomes.

Method Validity and Appropriateness

Both modeling approaches were rigorously tested and validated. Inputs were aligned with ZIP-code granularity, and output distributions were evaluated through spatial mapping and demand-capacity delta checks. Weekly reviews with GM stakeholders helped fine-tune weightings and reflect real-world deployment constraints. Sensitivity analysis confirmed that the model consistently selected 63–74 ZIP codes, even under changes in feature weighting and sparsity settings. This confirmed the model's stability and reliability for operational use.

Selected Method and Justification

The Greedy Optimization Algorithm was ultimately selected as the preferred deployment method due to its:

- Strong alignment with 74% of MIP results
- Operational speed and transparency for business users
- Scalability across states or future deployments
- Ability to prioritize high-demand, underserved ZIP codes effectively
- Ease of integration into GM's existing data infrastructure and dashboard

It delivered a fast, interpretable, and business-aligned solution while keeping implementation costs low—making it the ideal choice under budget-constrained, high-impact rollout scenarios.

Theoretical Relevance

The greedy method is grounded in local maximization theory, a subset of resource allocation problems where assets (chargers) are deployed one unit at a time to

maximize marginal utility. While it does not guarantee a globally optimal solution, its performance—validated against MIP—proves its high practical utility. For discrete deployments like charger rollouts, where ROI is tightly linked to placement efficiency, greedy algorithms strike a balance between speed and performance.

Key Limitations

Despite its strengths, the model carries some limitations:

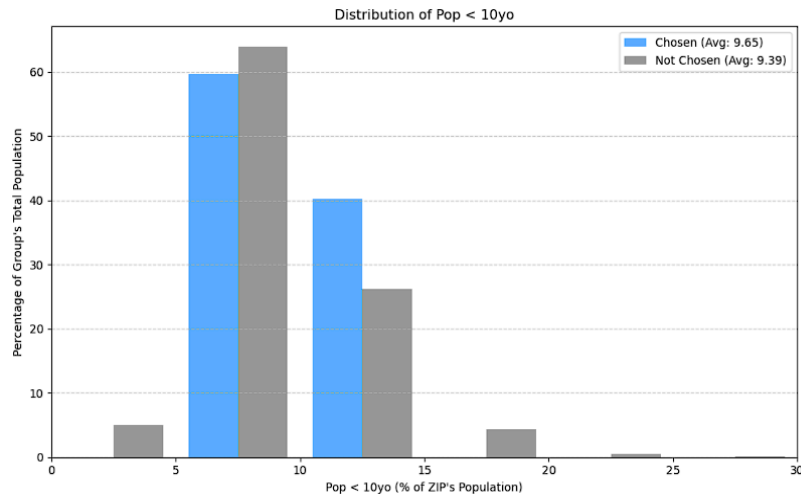
- Greedy optimization may miss globally optimal distributions
- Revenue and rent projections were based on static proxies, which may have underrepresented local economic variation
- Regression modeling, used earlier to explore EV adoption factors, was more diagnostic than predictive

However, these limitations were mitigated through regular validation, stakeholder feedback, and scenario testing, ensuring a deployment plan that is both actionable and reliable.

Demographic Drivers of Charger Placement

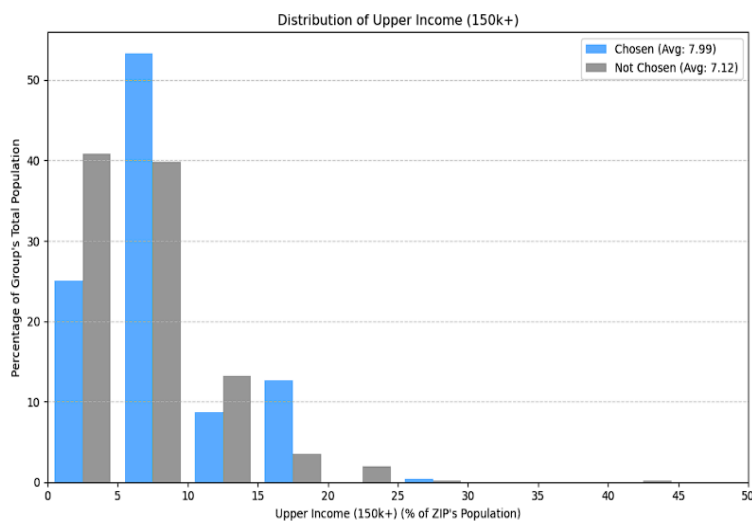
Demographic patterns play a critical role in understanding electric vehicle (EV) adoption and informing charger deployment strategies. Our analysis explored differences between ZIP codes selected for charger placements and those that were not, across four key dimensions: family structure and income levels, population density and education attainment, EV registration rates, and commute times. These insights helped us validate that charger locations were influenced by meaningful demand and socio-economic readiness indicators rather than random distribution.

Distribution of Age <10 Years:



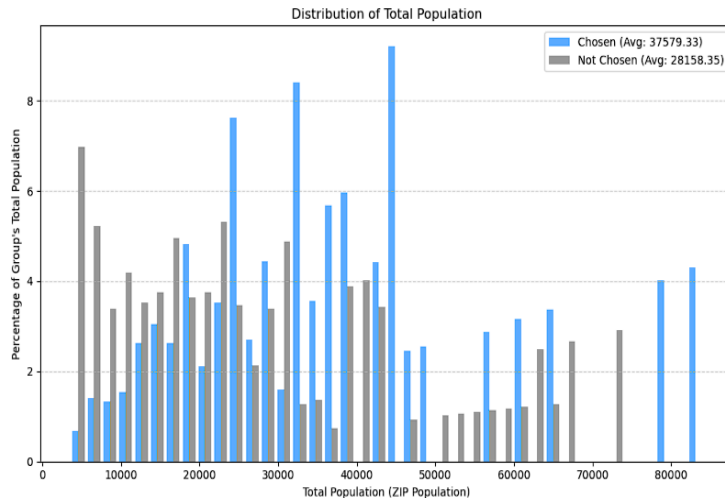
This compares the proportion of children under 10 years old across selected and non-selected ZIP codes. Results show that ZIPs with charger placements had a slightly higher percentage of young children (9.65% vs. 9.39%). This subtle difference suggests that family-oriented communities may exhibit patterns of early EV adoption, potentially linked to needs for reliable multi-vehicle households and regular commuting routines that support EV viability.

Distribution of Upper Income (\$150k+)



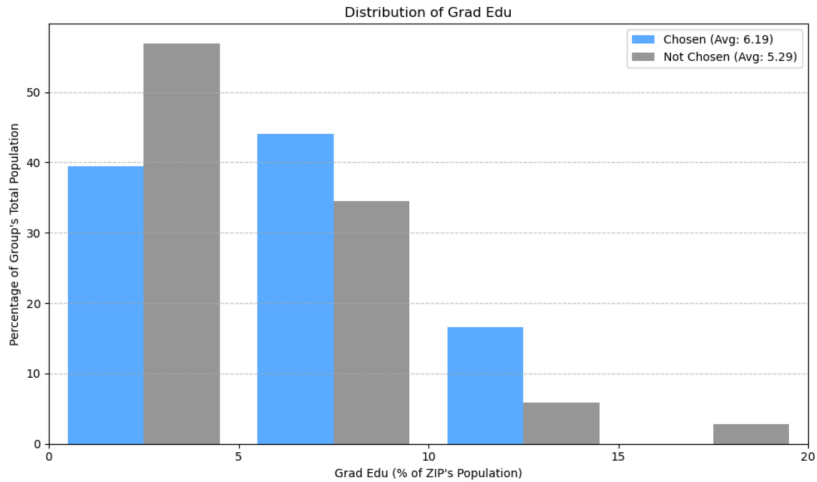
Here, we examine the share of residents earning over \$150,000 annually. Both selected and non-selected ZIPs showed similar income distributions, indicating that charger placement was not heavily biased toward high-income areas. This finding highlights that the strategy remained equitable, focusing more on accessibility, demand gaps, and potential charger usage rather than skewing toward wealthier neighborhoods.

Distribution of Total Population



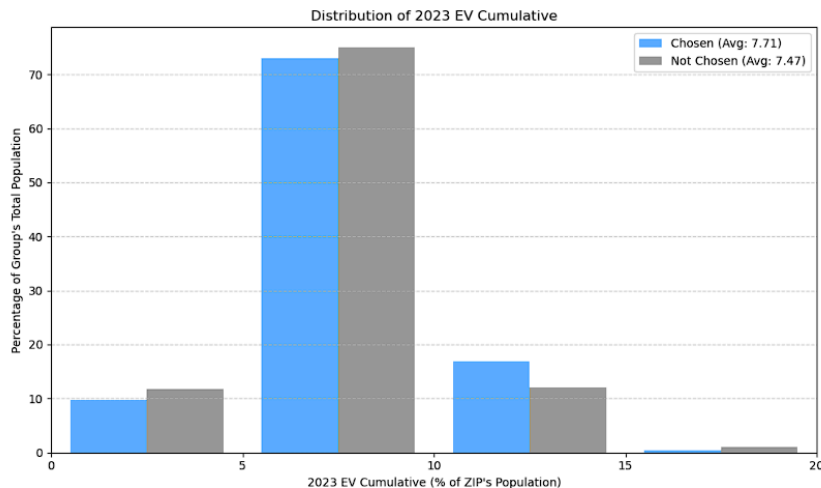
This graph illustrates differences in total population across ZIP codes. Charger placements favored more densely populated areas, with the average chosen ZIP having around 37,579 residents compared to 28,158 in non-chosen ZIPs. Targeting higher population centers ensures that chargers are placed where they are most likely to be utilized, aligning infrastructure investment with maximum usage potential.

Distribution of Graduate Education Levels



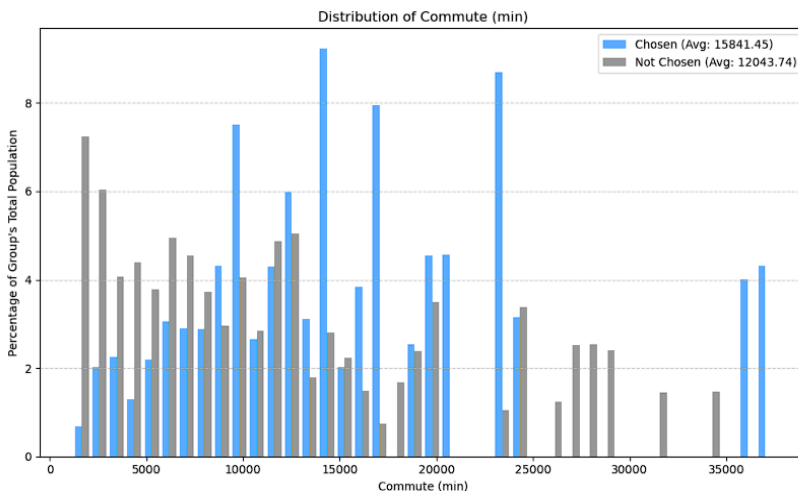
This chart analyzes education attainment, specifically the percentage of the population with graduate-level education. Chosen ZIP codes had a broader, more even spread across education levels, whereas non-chosen ZIPs were more concentrated at lower education levels. Although education level alone was not a primary driver, areas with slightly higher education levels often correlated with higher EV awareness and adoption readiness, indirectly supporting infrastructure demand.

Distribution of 2023 EV Cumulative Registrations



This compares cumulative EV registrations in 2023 between selected and non-selected ZIP codes. Both sets of ZIPs showed similar EV adoption rates, typically within the 6–9% range. This indicates that existing EV penetration was a basic threshold for consideration but was not the sole basis for charger placement. Instead, the strategy emphasized projecting future demand and filling infrastructure gaps beyond current ownership patterns.

Distribution of Commute Times

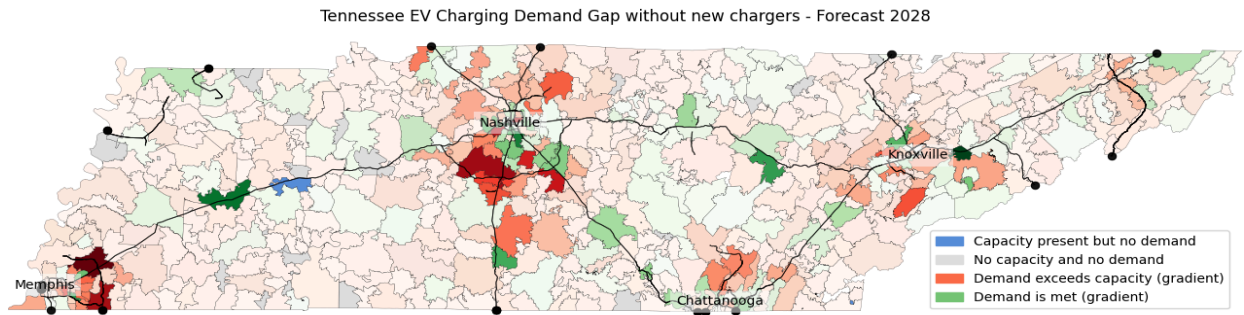


The final graph looks at commute times across ZIP codes. Areas selected for charger deployment generally had longer average commute times compared to those not selected. Longer commutes increase the likelihood of range anxiety and reliance on public chargers, making commute patterns a critical factor in maximizing charger usage rates and ensuring high return on infrastructure investment.

Business-Focused Findings

The ultimate goal of our analytical pipeline was to close the anticipated EV charging demand gap in Tennessee by 2028, ensuring that infrastructure deployment supports and accelerates General Motors' (GM) EV adoption strategy. By modeling the demand-supply dynamics at the ZIP-code level, we created a clear visual representation of both the problem and the solution. The following maps illustrate the transformation achieved through our optimized charger placement strategy—highlighting the direct business impact of our recommendations.

Tennessee EV Charging Demand Gap Without New Chargers (Forecast 2028)



The first map projects the EV charging landscape in Tennessee by 2028 if no additional chargers are deployed. Areas are color-coded to reflect the degree of unmet demand:

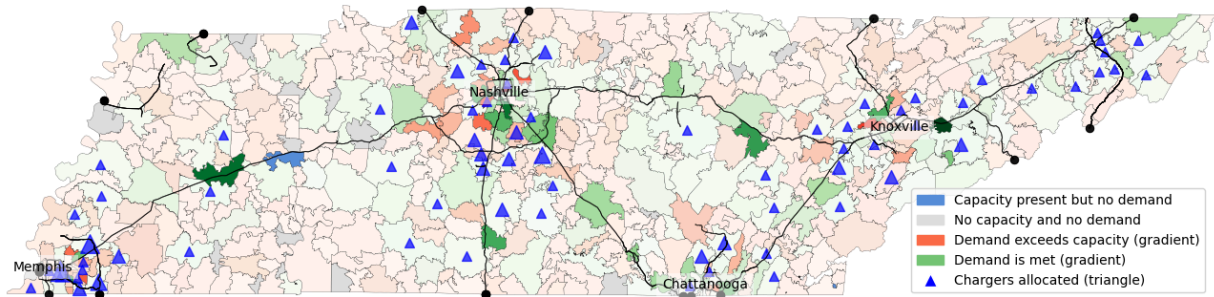
- Dark red and orange regions indicate zones where EV charging demand significantly exceeds existing infrastructure capacity.
- Light green zones represent areas where demand and supply are roughly balanced.
- Dark green patches show regions where capacity is present, but demand remains minimal.

This map makes it evident that major metropolitan areas—such as Nashville, Memphis, Knoxville, and Chattanooga—would experience critical infrastructure shortfalls without intervention. High-demand suburban corridors surrounding these urban centers are also at risk, showing deepening shades of red that signal future bottlenecks. Conversely, rural areas remain relatively under-pressured, but some isolated pockets show early indicators of emerging demand.

From a business standpoint, the map validates the strategic urgency: without proactive investment, GM risks losing potential EV customers to competitors due to infrastructure inaccessibility, customer dissatisfaction, and missed sales opportunities in rapidly growing markets.

Tennessee EV Charging Demand Gap With New Chargers (Forecast 2028)

Tennessee EV Charging Demand Gap with new chargers - Forecast 2028



The second map demonstrates the impact after deploying 100 new DC fast chargers, strategically placed according to the composite scoring model developed through our project. In this visualization:

- Blue triangles represent the newly allocated charger sites.
- The overall prevalence of red and orange zones is dramatically reduced, with most high-priority demand clusters either eliminated or significantly softened.
- Green zones (where demand is met) become more dominant across the state.

By inserting chargers into underserved but high-growth ZIP codes, the infrastructure landscape becomes substantially more balanced. Notably, areas around Nashville and Memphis—previously high-risk zones—show clear improvements, with demand-supply gaps either fully closed or markedly reduced. Emerging secondary cities and suburban growth areas also receive targeted support, creating a more equitable and future-ready infrastructure network.

Critically, the model achieved this outcome while adhering to GM's operational constraints—optimizing for utilization, demographic alignment, and long-term ROI. Additionally, sensitivity analyses confirmed that 63–74 ZIP codes consistently captured the highest composite scores, reinforcing the robustness of the allocation strategy.

From a business perspective, this optimized deployment ensures:

- Higher EV adoption rates by reducing range anxiety
- Better customer satisfaction and loyalty in key markets
- Stronger competitive positioning against EV rivals in Tennessee

- Alignment with federal and state incentives promoting public charging infrastructure

Strategic Implications

The transition from the "before" to the "after" map not only highlights the success of the modeling approach but also demonstrates GM's commitment to data-driven decision-making in building the EV ecosystem. Through thoughtful placement, GM can solidify its market share leadership, improve the economics of charger investments, and advance its vision of a zero-emissions future—all while supporting equitable access across Tennessee's diverse communities.

Execution Plan

Following the identification of optimal ZIP codes for DC fast charger placement, it is critical to translate strategic insights into effective, real-world action. Our execution plan outlines a phased and data-driven approach that prioritizes learning, scalability, and continuous improvement to maximize infrastructure ROI and customer adoption.

1. Launch Pilot in High-Priority ZIPs

The initial step involves launching a pilot deployment in a select group of high-priority ZIP codes identified through our optimization model. These sites will serve as early testing grounds to:

- Track charger profitability and return on investment (ROI)
- Measure real-world EV adoption trends and charger utilization
- Analyze user behavior, including dwell times, charging frequency, and patterns of charger congestion
- Gather operational insights related to charger maintenance, reliability, and site-specific challenges

Real-time dashboards will be developed to monitor key performance indicators (KPIs), enabling GM to rapidly detect trends, anomalies, and opportunities for improvement. This iterative feedback loop will ensure that early learnings directly inform broader rollout strategies.

2. Phased Rollout and Scaling

After validating success metrics during the pilot phase, the project will move into a phased expansion. The first scale-up wave will focus on deploying chargers across the top 50 ZIP codes based on pilot results and updated data feedback. KPIs such as usage rates, revenue per charger, downtime, and customer satisfaction scores will be continually assessed to refine placement and operations.

Once validated, the model will be extended to the next 50 ZIPs, followed by statewide and eventually cross-state deployments. This phased approach allows GM to:

- Systematically scale investments while minimizing risk
- Adjust deployment strategies based on real-world results
- Prioritize emerging high-demand areas dynamically
- Leverage economies of scale as procurement, installation, and maintenance operations mature

By following a controlled and data-backed rollout, GM can balance aggressive growth ambitions with operational excellence and risk management.

3. Replace Estimates with Real-World Data

A key pillar of the execution plan is to evolve the initial model by progressively integrating real-world performance data instead of relying solely on forecasted estimates. This involves:

- Incorporating actual charger usage and traffic volume data

- Updating demographic assumptions based on observed changes (e.g., migration trends, EV market penetration)
- Adjusting demand projections based on seasonality, regional economic activity, and new EV releases

By grounding future decisions in observed rather than predicted data, GM can achieve greater model accuracy, faster optimization cycles, and better allocation of financial and operational resources. This continuous learning model will ensure that charger placements remain closely aligned with evolving customer needs and market dynamics, securing GM's long-term leadership in the EV infrastructure space.

Next Steps

While the deployment plan and execution strategy provide a strong initial foundation, sustained success will require deeper data refinement, broader ecosystem engagement, market activation, and funding optimization. By combining deeper data insights, strategic partnerships, targeted marketing, and external funding, GM can not only accelerate EV infrastructure deployment but also create a scalable, sustainable competitive advantage in the rapidly growing electric mobility market. The following next steps outline the key initiatives to extend impact, maximize infrastructure ROI, and support long-term EV adoption growth.

1. Drill Down Beyond ZIP-Level Insights

The first refinement opportunity is to move from ZIP-code level planning to even more granular, street-level or neighborhood-specific analysis. While ZIP codes provide a strong macro framework for demand forecasting, micro-targeting at the block or traffic corridor level will allow GM to fine-tune charger placements for even greater utilization. Incorporating data such as localized traffic flows, commercial clustering, residential density, and commuter hotspots can dramatically increase placement precision and real-world effectiveness.

2. Explore Strategic Partnerships

Maximizing the value of each charging installation requires strategic partnerships beyond infrastructure deployment. Future success depends on partnering with grocery chains, gas stations, parking garages, malls, and other high-traffic site operators. These

collaborations can provide easier site access, lower installation costs, shared customer acquisition opportunities, and even co-marketing benefits. By embedding chargers into everyday destinations, GM can enhance convenience, increase charger visibility, and drive higher rates of repeat usage.

3. Launch Awareness-Focused Marketing Efforts

Merely deploying chargers is not sufficient—customers must be aware of their presence and benefits. GM should initiate localized marketing campaigns to boost charger visibility and accelerate EV adoption. Tactics could include digital screens at charger locations, banners, targeted social media ads, EV incentives at partner sites, and participation in local events promoting clean mobility. These efforts will help build trust, encourage trial, and normalize EV charging behaviors among a broader consumer base, driving faster infrastructure ROI.

4. Pursue Funding & Incentive Opportunities

To strengthen the business case and de-risk investments, GM should aggressively pursue external funding and incentive programs. Potential sources include:

- Federal and state green infrastructure grants
- Corporate ESG (Environmental, Social, Governance) partnerships
- Utility-sponsored EV rebate programs
- Private capital interested in sustainable mobility initiatives

By leveraging outside funding and reducing upfront capital exposure, GM can scale faster, improve deployment economics, and further align with broader governmental and societal sustainability goals.

Conclusion

The future of electric vehicle (EV) mobility hinges not just on the availability of compelling products, but on the supporting infrastructure that enables mass adoption. Through this project, General Motors (GM) has demonstrated a bold, data-driven approach to solving one of the most critical challenges in the EV transition: public charger deployment.

By analyzing Tennessee's EV market growth trends, demographic profiles, infrastructure gaps, and behavioral drivers, our team created a predictive, scalable, and operationally viable charger deployment strategy. Using a composite scoring framework and a combination of greedy optimization and integer programming validation, we prioritized 74 ZIP codes that will fulfill over 52% of forecasted EV charging demand by 2028 — ensuring high charger utilization, equitable access, and alignment with GM's zero-zero-zero vision.

Beyond technical success, the project created a clear execution blueprint: starting with pilots in high-priority ZIP codes, scaling deployment through phased rollouts, and progressively refining models with real-world data. Equally importantly, we outlined strategic next steps—drilling deeper into street-level insights, forming value-driven partnerships, launching awareness campaigns, and tapping into incentive programs—to ensure both operational success and business leadership.

Ultimately, this project transforms charger deployment from a reactive, fragmented process into a proactive, strategic asset for GM. It strengthens GM's position in the EV ecosystem, enhances brand loyalty through improved customer experience, and sets a replicable model for future nationwide rollouts. Through disciplined execution and continuous learning, GM can accelerate EV adoption not just in Tennessee, but across the broader United States market, shaping the future of sustainable transportation.

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